

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda (UTC+3)

PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years building real-time Node.js + TypeScript systems on AWS and GCP. At MTailor I migrated 20+ production apps from Parse/MongoDB to Firebase with zero downtime, cutting infrastructure spend by \$5,000/month while keeping the live system running. Built a real-time two-way MongoDB↔Firestore sync via Google Pub/Sub (event-driven, WebSocket-style throughput) and ran daily 2M+ record jobs at Dr Wealth keeping customer prices fresh. Now applying that real-time, high-throughput backend toolkit to a Senior Backend role at Gamdom.

CORE COMPETENCIES

Node.js + TypeScript

PostgreSQL & Schema Design

AWS (EC2, S3, EBS, IAM)

Real-Time Systems & WebSockets

High-Throughput Pipelines

Microservices (Kafka, Docker, K8s)

Zero-Downtime Migrations

Event-Driven Architecture (Pub/Sub)

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Built a **real-time two-way sync** between MongoDB and Firestore using **Node.js** and **Google Pub/Sub**, keeping both stores consistent during a multi-month live migration — event-driven, WebSocket-style throughput analogous to socket.io patterns.
- Led **zero-downtime migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP, reporting directly to the CTO.
- Saved the company **\$5,000/month** by migrating all services off **AWS** (EC2, S3, EBS) to GCP.
- Migrated file storage from **Amazon S3 to GCS** via a Python script with checksum verification.
- Implemented Express Shipping API + UI end-to-end, generating an additional **\$40 revenue per order**.
- Built a 3D visualisation feature for customers using video overlay and ffmpeg, increasing buyer conversion.

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Hired via Upwork to execute a complex backend migration from Parse/MongoDB to Firebase Firestore on GCP.
- Built a migration script reading from MongoDB and writing to Firestore with complex processing logic and schema transforms.
- Wrote engineer-facing documentation on the new Firebase SDKs (Node.js, Python) and trained the Ops team on the new dashboard and backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Built ad-hoc jobs querying over **2 million Firestore records** daily, keeping customer prices current from Morningstar APIs — high-throughput batched updates.
- Extended backend APIs using **Firebase Cloud Functions** and Express.js on Heroku.
- Added responsive UI pages to the PWA using React, Tailwind, and HTML.

CodeBits

Jan 2020 – Jul 2021

Software Engineer · Uganda Remote

- Architected a **microservices backend** for FIDA Uganda's case management system using **Apache Kafka, Docker, and Kubernetes** — async messaging architecture for real-time event distribution.
- Led a team of **4 developers** building legal-tech systems for NGOs in Uganda.
- Built a USSD service for legal-aid providers communicating with a Node.js backend via gRPC.
- Developed cross-platform mobile app for LASPNET using React Native + Expo with Firebase push notifications.

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Full-stack development for a US-based agency, reporting directly to the CTO.
- Built backends for DeFi applications using Web3 and Node.js.

PROJECTS

Parse → Firebase Migration (MTailor) Hero

Migrated 20+ production apps with zero downtime, cutting \$5,000/month in infrastructure costs.

Node.js · Firebase · GCP · MongoDB · Pub/Sub · AWS S3/EC2

Real-Time MongoDB ↔ Firestore Sync Real-time

Bidirectional Pub/Sub pipeline supporting parallel production traffic during migration — event-driven, WebSocket-style architecture.

Node.js · Google Pub/Sub · MongoDB · Firestore

FIDA Uganda Microservices Backend Architecture

Microservices case-management system with async messaging — Kafka topics for event distribution across services.

Node.js · Apache Kafka · Docker · Kubernetes · gRPC · NATS Streaming

Dr Wealth Real-Time Stock Prices High-throughput

Daily jobs querying 2M+ Firestore records to refresh customer prices from Morningstar APIs.

Node.js · Firestore · Firebase Cloud Functions · Express.js

EDUCATION

BSc Computer Engineering — Makerere University, Kampala, Uganda

Aug 2018 – Sep 2022

SKILLS

Proficient: Node.js, TypeScript, JavaScript, Firebase, React, GCP, Python, AWS (EC2, S3, EBS), MongoDB, SQL, Git, Linux, HTML/CSS

Intermediate: PostgreSQL, Go, Rust, React Native, Apache Kafka, Docker, Kubernetes **Languages:** English, Kiswahili